Optimal Market Making Under Inventory Risk:

A Dissection of the Avellaneda-Stoikov Simulation

Theory, Implementation, and Parameter Optimisation

Handiko Gesang, based on Avellaneda & Stoikov (2008)

Abstract

This essay provides a comprehensive technical and theoretical analysis of an interactive market-making simulator built around the Avellaneda-Stoikov (2008) model. We examine the stochastic control problem that motivates the model, derive the reservation price and optimal spread formulas, explain how Geometric Brownian Motion and Poisson arrival processes are implemented in the simulation code, and offer systematic guidance for parameter selection. The essay concludes with a discussion of the conditions under which a market maker maximises risk-adjusted profitability, with particular emphasis on the tension between spread revenue, inventory risk, and adverse selection.

1. Introduction

Market making is one of the oldest and most essential functions in organised financial markets. A market maker simultaneously posts a *bid* price at which she is willing to buy an asset, and an *ask* price at which she is willing to sell the same asset. The difference between the two, the *bid-ask spread*, is the primary source of her revenue. In exchange for providing this continuous two-sided liquidity, the market maker bears two distinct risks: *inventory risk* — the danger that an accumulating directional position will lose value as the price moves against her — and *adverse selection risk* — the possibility that informed traders will systematically take the better side of her quotes, leaving her with an unfavourable position.

For decades, the canonical treatment of these problems consisted of two largely separate literatures. The inventory-management tradition, following Ho and Stoll (1981), modelled the market maker as a utility-maximising agent who adjusts her quotes to control the speed at which inventory reverts to zero. The information-based tradition, following Glosten and Milgrom (1985) and Kyle (1985), focused on how the market maker updates prices in response to the information content of order flow. A unified and tractable

framework reconciling these two perspectives arrived in 2008, when Marco Avellaneda and Sasha Stoikov published their seminal paper, *"High-frequency trading in a limit order book."*

The Avellaneda-Stoikov (A-S) model casts market making as a continuous-time stochastic control problem. It derives, in closed form, the exact bid and ask prices that maximise the market maker's expected utility of terminal wealth. The result is elegant: the optimal quotes are expressed as symmetric adjustments around a *reservation price* that is itself biased away from the mid-market price by an amount proportional to the current inventory. This essay dissects every layer of the resulting simulation — from the continuous-time mathematics that motivates the formulas, to the JavaScript code that implements them in real time, to the practical guidance that allows a user to navigate the parameter space toward consistently profitable strategies.

2. Theoretical Foundations

2.1 The Price Process: Geometric Brownian Motion

The mid-market price $S(t)$ is assumed to follow Geometric Brownian Motion (GBM):

$$dS = \mu S\, dt + \sigma S\, dW_t$$

where W_t is a standard Wiener process, μ is the drift, and $\sigma > 0$ is the volatility. In the simulation, μ is set to zero, reflecting the assumption that the market maker has no directional view. By applying Itô's lemma to $\log S$, the exact discrete-time update rule becomes:

$$S(t + \Delta t) = S(t) \cdot \exp(-\sigma^2 \Delta t/2 + \sigma\sqrt{\Delta t} \cdot Z)$$

where $Z \sim N(0, 1)$ is a standard normal draw. The term $-\sigma^2\Delta t/2$ is the Itô correction that keeps the process drift-neutral in levels ($E[S(t+\Delta t)] = S(t)$ when $\mu = 0$), and $\sigma\sqrt{\Delta t}$ scales the random shock correctly for the time increment. In the code, this appears as:

```
S = S * Math.exp(-0.5 * σ² * DT  +  σ * √DT * randn())
```

where `DT = 0.004` is the model time increment per simulation tick, and `randn()` generates a standard normal sample via the Box-Muller transform. The choice of GBM is deliberate: it ensures prices remain positive, produces the log-normal return distribution that underlies the Black-Scholes framework that Avellaneda and Stoikov also invoke, and has the desirable property that percentage volatility σ is constant through time.

2.2 The Stochastic Control Problem

The market maker seeks to choose, at each instant t, the bid price $P^b(t)$ and ask price $P^a(t)$ to maximise expected exponential utility of terminal wealth at horizon T:

$$\max \quad E[-\exp(-\gamma (X_T + q_T \cdot S_T))]$$

subject to the dynamics of cash $X(t)$ and inventory $q(t)$. Here $\gamma > 0$ is the absolute risk aversion coefficient, $q_T \cdot S_T$ is the mark-to-market value of any remaining inventory, and the exponential utility function penalises losses more severely than it rewards equivalent gains — a preference structure consistent with the practice of professional risk management.

Trades arrive as a doubly stochastic Poisson process. The intensity at which sell orders arrive (hitting the bid) depends on how far the bid is placed below the mid-market price. Specifically, Avellaneda and Stoikov assume an exponential intensity function:

$$\lambda^b = A \cdot \exp(-k \cdot \delta^b) \qquad \lambda^a = A \cdot \exp(-k \cdot \delta^a)$$

where $\delta^b = S - P^b$ is the distance of the bid from mid, $\delta^a = P^a - S$ is the distance of the ask from mid, $A > 0$ is the base arrival rate, and $k > 0$ governs how rapidly arrival intensity decays as quotes are moved away from mid. This functional form captures a crucial feature of real limit order books: the further away a quote is posted, the less likely it is to be filled, because fewer counterparties are willing to transact at a price far from fair value.

2.3 The Hamilton-Jacobi-Bellman Equation and Its Solution

Avellaneda and Stoikov solve the control problem via the Hamilton-Jacobi-Bellman (HJB) equation. The key insight is that the value function — the maximum achievable expected utility as a function of current state (t, x, q, S) — admits a transformation that decouples the problem into a tractable ordinary differential equation. After an ansatz of the form

$$u(t, x, q, S) = -\exp(-\gamma (x + q \cdot S) \cdot \theta(t, q))$$

the problem reduces to finding $\theta(t, q)$, and eventually yields two closed-form results that are the heart of the model.

First, the **reservation price** — the price at which the market maker, given her current inventory, is indifferent between buying and selling — is:

$$r(S, q, t) = S - q \cdot \gamma \cdot \sigma^2 \cdot (T - t)$$

This formula is profound in its simplicity. When $q = 0$, the reservation price equals the mid-market price. When $q > 0$ (the market maker is long), $r < S$: she values the asset below mid because she already has too much of it and bears downside inventory risk. When $q < 0$ (short), $r > S$. The magnitude of this adjustment grows with risk aversion γ, variance σ^2, and the remaining time horizon $T - t$ — all of which sensibly increase the urgency of inventory liquidation.

Second, the **optimal spread** — the total width of the quote — is:

$$\delta^* = \gamma \cdot \sigma^2 \cdot (T - t) + (2/\gamma) \cdot \ln(1 + \gamma/k)$$

The first term, $\gamma\sigma^2(T-t)$, is the **inventory risk premium**. It reflects the cost of holding a risky position over the remaining horizon: the higher the volatility and risk aversion, the wider the market maker must quote to compensate herself for bearing inventory risk. The second term, $(2/\gamma) \cdot \ln(1 + \gamma/k)$, is the **adverse selection premium**. It accounts for the

order book depth parameter *k*: when *k* is large, order flow is sensitive to price distance, so the market maker can afford to post tighter spreads without significantly altering her fill probability. The optimal bid and ask are then placed symmetrically around the reservation price:

$$P^b = r - \delta^*/2 + skew$$

$$P^a = r + \delta^*/2 + skew$$

The *skew* term is a manual override not present in the original A-S paper. It allows the user to shift both quotes uniformly up or down, reflecting a directional view layered on top of the model's inventory-optimal positioning.

3. The Poisson Trade Arrival Model

3.1 Continuous-Time Arrivals

In continuous time, the probability that at least one order arrives in the infinitesimal interval *[t, t+dt]* for a Poisson process with intensity *λ* is:

$$P(\text{at least one arrival in dt}) = 1 - \exp(-\lambda \cdot dt) \approx \lambda \cdot dt \quad \text{for small dt}$$

The intensity itself depends on quote placement. For the bid side, the market maker receives sell orders (she buys) at rate:

$$\lambda^b = A \cdot \exp(-k \cdot \max(0, S - P^b))$$

This says that when the bid *P^b* equals the mid-market price *S*, the arrival rate is at its maximum *A*. Every unit of distance the bid moves below mid reduces the intensity multiplicatively by the factor *exp(-k)* per unit. The `max(0, ...)` clamp prevents the formula from *increasing* intensity when quotes are placed inside the spread — a situation that does not arise under optimal A-S quoting but that must be handled defensively in simulation code.

3.2 Discrete Approximation in the Simulation

Because the simulation runs in discrete time steps of `DT = 0.004` model units per 160ms wall-clock tick, the continuous-time probability is approximated by the exact Poisson CDF evaluated at the discrete increment:

$$P_fill = 1 - \exp(-A \cdot \exp(-k \cdot dist) \cdot DT)$$

In the code, this is the `poissonProb()` function, which is called independently for the bid side (`dist = S - bid`) and the ask side (`dist = ask - S`). A uniform draw on *[0, 1]* is then compared against each probability to determine whether a fill occurs on that side in that tick. Note that both sides can fill simultaneously in the same tick — representing a

scenario where the market maker both buys and sells within a single interval, a simplification that becomes negligible for sufficiently small `DT`.

Each fill immediately updates the state: a bid fill decrements cash by *P^b* and increments inventory by 1; an ask fill increments cash by *P^a* and decrements inventory by 1. The mark-to-market profit and loss at any point is:

$$\mathbf{PnL(t) = X(t) + q(t) \cdot S(t)}$$

where *X(t)* is cumulative cash and both are initialised to zero. This quantity, also called the equity curve, is what appears in the lower-right chart in the simulation — and because it is never rolled or truncated, it presents the complete history of the strategy's performance from session start.

4. Code Architecture and Implementation

4.1 Overall Structure

The simulation is implemented as a single self-contained HTML file with no build toolchain or server-side dependencies. All logic runs in the browser via vanilla JavaScript, with Chart.js (loaded from a CDN) providing the two real-time chart surfaces. The code is structured around five conceptual layers:

1. **State variables:** S (mid price), cash, inv (inventory), fills, tick — the complete simulation state at each moment.
2. **Price engine:** The GBM step computed inside simStep(), producing a new S each tick.
3. **Quote engine:** The asQuotes() function implementing the A-S reservation price and optimal spread.
4. **Fill engine:** The poissonProb() function and the two Bernoulli trials per tick that govern order execution.
5. **Rendering layer:** pushData(), updateMetrics(), and addLog(), which translate simulation state into chart updates, metric cards, and the fill log.

The simulation loop is driven by `setInterval(simStep, 160)`, which fires every 160ms of wall-clock time. Each call to `simStep()` advances the model by `DT = 0.004` units of model time, corresponding to roughly 4 seconds at an annualised clock if one interprets *T = 1* as one trading year of 252 days.

4.2 The asQuotes() Function

The core of the simulation is the quote calculation function, reproduced below in annotated pseudocode:

```
function asQuotes(S, q, params):
  1.  r  = S - q * γ * σ² * T                // reservation price
```

```
2.  hs = γ*σ²*T/2 + (1/γ)*ln(1 + γ/k)   // half-spread δ*/2
3.  bid  = r - hs + skew
4.  ask  = r + hs + skew
```

Step 1 implements the inventory adjustment: as q grows positive (long inventory), r is pulled below S, causing both bid and ask to shift downward and making the ask more competitive — the model is passively selling into strength to unwind the long. The converse applies when q is negative. This *inventory skew* is the central risk-management mechanism of the A-S model.

Step 2 computes the optimal half-spread. Note the $T - t$ term is simplified in the implementation to a fixed parameter T, making the model stationary rather than time-decaying. This is a common practical adaptation: in live HFT applications, T is best interpreted as a "risk horizon" or "inventory decay timescale" rather than a literal time to close.

4.3 Random Number Generation

The normal samples required for GBM are produced by the Box-Muller transform, implemented as the `randn()` function. Given two independent uniform samples $U_1, U_2 \sim Uniform(0, 1)$, the transform yields:

$$Z = \sqrt{(-2 \ln U_1)} \cdot \cos(2\pi U_2) \sim N(0, 1)$$

The while loops guarding against `u = 0` are required because `Math.log(0)` is undefined. In practice, the probability of exactly drawing zero from a floating-point uniform is astronomically small but nonzero, so the guard is mathematically necessary.

4.4 Chart Architecture

Two Chart.js line charts are maintained in parallel. The **Price/Bid/Ask chart** uses a rolling window of `MAX = 200` ticks: when the array reaches capacity, the oldest data point is shifted off the front before the newest is pushed to the back. This keeps the chart focused on the recent past and prevents memory growth. The **PnL chart**, by contrast, is never truncated — data is only ever appended. This design choice means the equity curve always displays the *entire session history*, allowing the user to assess long-run performance at a glance. Both charts are updated with the `'none'` animation mode, which redraws the canvas immediately without transitions — essential for the visual responsiveness of a real-time simulation running at approximately 6 ticks per second.

5. Parameter Guide: What Each Control Does

The simulator exposes six parameters that together span the full A-S model. Understanding each one precisely is the prerequisite for informed strategy design.

Parameter	Range	Default	Effect on Model
Risk aversion γ	0.01 – 1.0	0.10	Controls both spread width and inventory skew. Higher γ widens the spread and makes quotes more aggressively retreat from inventory positions.
Volatility σ	0.05 – 0.5	0.15	Drives both the price randomness (GBM diffusion) and the spread formula. Higher σ increases the price chart turbulence and forces wider quotes.
Order depth k	0.1 – 5.0	1.50	Controls how sensitively fill probability decays with spread distance. High k = fills collapse quickly with distance; tighter spreads required for activity.
Trade frequency A	0.5 – 20	5.0	The base Poisson arrival intensity. Represents taker aggression. High A fills the log rapidly; low A produces a quiet book with rare executions.
Manual skew	−3 to +3	0.0	Shifts reservation price, and therefore both quotes, by a fixed amount. Positive = lean bullish (quote higher). Overrides the model's neutral stance.
Risk horizon T	0.05 – 2.0	0.50	The timescale over which inventory risk is measured. Longer T amplifies the inventory skew and spread premium; shorter T makes the model more permissive.

5.1 Risk Aversion γ in Depth

The risk aversion parameter γ is the most consequential single dial in the model. It appears in three places in the formulas: as a direct multiplier in the inventory adjustment of the reservation price ($-q\gamma\sigma^2T$), as a coefficient in the inventory-risk term of the spread ($\gamma\sigma^2T/2$), and as the denominator in the adverse-selection term ($(1/\gamma)\cdot ln(1 + \gamma/k)$). The last relationship is non-monotone: at very low γ, the adverse-selection spread component diverges as $(1/\gamma)\cdot ln(1 + \gamma/k) \rightarrow k/\gamma^2$ for small γ. This means extremely risk-tolerant market makers actually set very wide spreads to compensate for the adverse selection they are no longer avoiding through inventory management. The sweet spot for narrow spreads is at intermediate γ

5.2 Order Depth k and the Fill Rate Trade-off

The parameter k governs the elasticity of order flow with respect to price distance. In a book with high k, traders are highly price-sensitive: even a small distance from mid sharply reduces the arrival rate. In a book with low k, traders are relatively insensitive to price and will transact even at wide quotes. The practical implication is that k encodes the *liquidity* of the market: deep, active markets have low k (fills are plentiful even at wide spreads),

while thin, illiquid markets have high k (fill probability collapses quickly away from mid). Optimal spreads are narrower in high-k environments because the adverse-selection spread component $(2/\gamma)\cdot ln(1 + \gamma/k)$ shrinks as k increases, reflecting that price-sensitive flow is less likely to be informed.

5.3 Manual Skew as a Directional Overlay

The manual skew is not part of the original A-S model and should be used with full awareness of its effect. A positive skew rigidly shifts both quotes upward: the ask moves further from mid (harder to fill), while the bid moves closer to mid (easier to fill). This creates a systematic *buying bias* — the market maker will accumulate long inventory over time. A negative skew has the opposite effect, creating a selling bias. This is appropriate when the user wishes to combine the A-S inventory management framework with a directional signal. However, if the price moves against the skew direction, losses can compound rapidly since neither the reservation price adjustment nor the spread width change in response to the skew. In production systems, skew is typically derived from an alpha signal rather than set manually.

6. Choosing Optimal Parameters for Maximum Risk-Adjusted PnL

There is no universally "optimal" parameter set because the profit landscape is determined by the trade-off between three quantities that cannot simultaneously be maximised: *spread revenue* (higher when spreads are wide), *fill rate* (higher when spreads are narrow), and *inventory risk* (lower when positions revert quickly). The following principles, derived from both the theory and empirical exploration of the simulator, guide the parameter selection process.

6.1 The Core PnL Decomposition

The expected PnL per unit time can be decomposed as follows. Each bid fill earns a gross profit equal to the mid-to-bid half-spread $\delta^*/2$ (since the asset is bought below mid and will eventually revert to mid under zero drift). The same applies to ask fills. Simultaneously, the market maker suffers a continuous *carry cost* from holding inventory: a position of size q in a market with instantaneous variance σ^2 generates a mark-to-market P&L variance of $q^2\sigma^2$

The A-S model sets the spread exactly wide enough so that the marginal fill revenue equals the marginal inventory cost at any inventory level. When the market maker follows the model's prescriptions without deviation, she earns, in expectation, the adverse-selection premium $(2/\gamma)\cdot ln(1 + \gamma/k)$ per unit of spread traded — a positive quantity for all valid parameter values.

6.2 Regime-Based Parameter Recommendations

The following recommendations are structured around three distinct market regimes, each of which rewards a different configuration:

Low-Volatility, Active Market

When the price chart shows tight oscillations and the fill log is active, the market maker is in the most favourable environment. The optimal configuration is:

- **γ = 0.05 – 0.15:** Low risk aversion keeps spreads narrow and maximises fill rate.
- **σ = 0.05 – 0.15:** Low volatility means low inventory carry cost, so larger positions can be tolerated.
- **k = 1.0 – 2.0:** Moderate depth; spreads are narrow enough to attract fills without giving away too much edge.
- **A = 5 – 10:** Moderate to high activity generates a steady stream of fills.
- **T = 0.3 – 0.5:** Short horizon keeps inventory adjustments responsive.

High-Volatility Market

When sigma is elevated ($\sigma > 0.3$), the inventory carry cost becomes punishing. The model automatically widens spreads, but the user should reinforce this with:

- **γ = 0.3 – 0.7:** Higher risk aversion triggers more aggressive inventory skewing, reducing the likelihood of large directional positions.
- **T = 0.1 – 0.3:** A short horizon makes the inventory correction very sharp: even small positions generate large quote skews.
- **A = 3 – 6:** Moderate activity. In high-vol conditions, aggressive fill rates can lead to large inventory swings faster than the model can unwind them.
- **Skew = 0:** In high-vol environments, any directional bet amplifies loss risk sharply. Keep the model symmetric.

Thin Market (High k, Low A)

When fills are infrequent (low A) and order flow is price-sensitive (high k), the market maker must tighten quotes to attract any volume at all:

- **γ = 0.05 – 0.1:** Low risk aversion keeps half-spreads narrow enough to generate fills.
- **A = 1 – 3:** Accept that fills will be rare; focus on quality of each fill over quantity.
- **k = 3 – 5:** High k means even small spread distances dramatically reduce fill probability. Spreads must be kept tight.
- **T = 0.5 – 1.0:** Longer horizon relaxes inventory urgency, appropriate when fills are slow.

6.3 The Inventory Leash Principle

Regardless of market regime, one heuristic holds universally: observe the inventory bar carefully. The A-S model is designed to mean-revert inventory through asymmetric quoting, but it can only do so passively — it relies on counterparty order flow to unwind positions. If inventory grows extreme (the bar reaches its limits) and the price is trending against the position, mark-to-market PnL will deteriorate rapidly despite all spread revenues earned.

The practical remedy is to increase γ and/or decrease T while the inventory is large. Both actions amplify the magnitude of the reservation price adjustment $-q\gamma\sigma^2T$, forcing the quotes further into the side of the market that would unwind the position. In a live trading context, this logic is typically automated: many production implementations make γ a function of $|q|$, so that risk aversion automatically escalates as inventory grows. The simulator does not implement this feedback loop, but the user can replicate its effect manually by adjusting the slider in response to inventory buildups.

6.4 The Spread-Volume Trade-off

A final principle worth quantifying: under the exponential intensity model, reducing the half-spread by Δ increases the fill rate by a factor of $exp(k \cdot \Delta)$. For $k = 1.5$ and $\Delta = 0.5$, this is a factor of $exp(0.75) \approx 2.1x$ — doubling the volume. However, each fill generates less revenue. The net expected profit rate from the bid side is proportional to:

$$\text{Revenue rate} \propto (\delta^*/2) \cdot A \cdot \exp(-k \cdot \delta^*/2)$$

This is maximised when $d/d(\delta^*/2)\ [\ (\delta^*/2) \cdot exp(-k \cdot \delta^*/2)\ ] = 0$, yielding $\delta^*/2 = 1/k$. This is the "naive" optimal half-spread from a pure revenue perspective. The A-S model produces a wider spread than $1/k$ because it also prices in the inventory risk premium and adverse selection cost. When $\gamma \cdot \sigma^2 \cdot T$ is small relative to $(2/\gamma) \cdot ln(1 + \gamma/k)$, the spread is driven primarily by adverse selection concerns and the optimal value is close to the pure revenue maximum of $2/k$.

7. Limitations and Possible Extensions

7.1 Model Limitations

The simulation, faithful to the original A-S paper, incorporates several simplifying assumptions that limit its fidelity to real markets. First, the price impact of the market maker's own trades is ignored: in reality, posting a very tight bid in a thin market will attract adverse-selection volume that pushes the price down. Second, the Poisson intensity is assumed to depend only on the distance from mid-market, not on the broader order book state, news flow, or time of day. Third, all fills are for a single unit of size: the model does not address partial fills, queuing position, or size-dependent adverse selection.

The stationary treatment of T deserves further mention. In the original paper, $T - t$ shrinks over the trading session, causing spreads to tighten and inventory urgency to increase as the close approaches. This end-of-day effect is well documented empirically. The

simulation's fixed *T* removes this dynamic, making it more appropriate as a model of a continuously rolling strategy than a single-session market maker.

7.2 Directions for Extension

Several natural extensions would bring the simulation closer to a research-grade tool:

1. **Stochastic volatility:** Replacing constant σ with a Heston or SABR process would produce the volatility clustering observed in real returns, significantly altering the spread dynamics.
2. **Asymmetric intensity:** Allowing A and k to differ between the bid and ask sides models a market with directional order imbalance, a more realistic depiction of most electronic books.
3. **Dynamic γ:** Implementing $\gamma = \gamma_0 + \lambda|q|$ makes risk aversion endogenous to inventory, closing the feedback loop that production systems rely on.
4. **Multiple assets:** Extending to a two-asset setting with correlation ρ allows study of cross-asset inventory hedging — a key feature of multi-currency or multi-leg market making.
5. **Time-varying T:** Restoring the original countdown timer would reproduce the end-of-session spread compression and add a session-management dimension to the strategy.

8. Conclusion

The Avellaneda-Stoikov model occupies a distinguished position in the market microstructure literature: it is simultaneously theoretically rigorous, analytically tractable, and practically applicable. The simulation presented here makes its mechanics directly observable, allowing a user to develop — through experimentation rather than abstraction — a visceral intuition for how inventory risk and spread revenue interact.

The key insight that should survive any level of model simplification is this: a market maker's profitability is not primarily determined by the width of her spread, but by how well she manages the risk she accumulates between fills. The A-S model addresses this directly by building inventory management into the quoting logic itself, rather than treating it as a separate post-hoc problem. The reservation price adjustment ensures that as inventory grows, quotes organically lean toward the unwinding side — not through a hard rule, but as the natural consequence of maximising expected utility.

For users seeking to maximise risk-adjusted PnL in the simulation, the most reliable approach is to calibrate *γ* and *T* to match the observed speed of price oscillation against the speed of fill arrival: if inventory is building faster than price is reverting, both parameters should be increased. If fills are too rare to sustain the strategy, *γ* should be reduced to narrow the spread and attract more counterparty flow. This iterative, regime-aware calibration is precisely what the six-parameter interface of the simulator is designed to support.

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